



VPF-T7R3-DSH-01 Case t7_r3 | 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Virtual design package - simulation caliber (no physical build)

VBF Technical Datasheet - HEV Battery Cell (virtual design)

****Document number**:** VBF-T7R3-DSH-01 · ****Case**:** t7_r3 · ****Date**:** 2026-08-26

All values are simulation/parameter-set sourced; caliber noted per row. Prepared / Reviewed / Approved:

Field | Value | Source

Rated capacity (Ah) | 6.28 Ah (nominal, set) - verified 6.2569 Ah at 1C discharge (SPMe); 6.28 Ah (DFN) | `r7\_v19\_final\_1c.json`, `r6\_v19\_1c\_dfn.json`

Nominal voltage / window | 3.7 V class; window 2.5 - 4.2 V (midpoint voltage at 1C: 4.047 V) | Chen2020 cut-offs; `r7\_v19\_final\_energy.json:midpoint\_voltage\_v`

Rated energy (Wh) | 22.491 Wh (intV·I■C dt, I■C = 6.28 A) | `r7\_v19\_final\_energy.json:energy\_wh`

Energy density | ****482.4 Wh/kg**** (contract caliber, electrolyte excluded); 417.9 Wh/kg incl. 7.21 g electrolyte; volumetric 970.7 Wh/L | `r7\_v19\_final\_energy.json` + electrolyte derivation (1.2 g/cm³ literature density)

Maximum continuous discharge rate | 1C (6.28 A) verified to 2.5 V; higher-rate discharge not simulated (5C protocol exists but not run - no criterion) | `r7\_v19\_final\_1c.json`

Fast-charge capability (4C) | 4C = 25.12 A: CC charge accepts 0.778 Ah (12.4 % SOC) before hitting 4.2 V; ****no lithium plating**** (anode surface potential min +0.0171 V, 0 negative instants); T_max during charge 363.2 K (90.1 °C, lumped thermal) | `r7\_v19\_final\_4c45c.json`

Operating temperature range | 25 °C (1C discharge, simulation default) and 45 °C (4C charge + aging) verified; below 0 °C models ****not simulated**** | protocols used

Cycle life | 100 cycles @ 45 °C (1C, SEI ec-reaction-limited model) simulated: SEI thickness end-of-test 512.9 nm (criterion ≤ 550 nm PASS); per-cycle net capacity entries ~0.23-0.26 Ah reflect the charge-imbalance metric of the aging model, ****not**** a cell-level usable-capacity retention curve - end-of-life retention extrapolation ****Not simulated**** | `r7\_v19\_final\_aging45c.json`

Safety determination | Nail penetration 10 W with designed cooling hA = 0.5 W/K: ****no thermal runaway**** (T_max 318.2 K); 4C charge: plating-free (above). Mandatory cooling requirement: hA ≥ 0.3 W/K | `r7\_v19\_final\_nail\_hA05.json`, sweep logs

Dimensions | 65 mm × 1580 mm unwound electrode strip; stack thickness 225.6 μm (excludes shell - not modeled) | Chen2020 geometry; `thickness\_m`

Mass | 46.62 g (electrolyte excluded, contract); 53.83 g incl. electrolyte | `r7\_v19\_final\_energy.json:mass\_kg` + derivation

DC internal resistance | 0.120 mohm (V(t■)-V(10 %)) / I■C | `r7\_v19\_final\_energy.json:dcr\_ohm`

****Customer-facing notes**:** Chemistry NMC811/"graphite with low-transference (t+ ~ 0.6) electrolyte formulation" - the exact salt/solvent recipe is a formulation estimate (see DS-01 Note 1), to be fixed by electrolyte qualification. All safety/performance claims above are simulation-caliber (virtual design); physical qualification (UN 38.3, cycle-life ext, thermal runaway physical) requires prototype build.